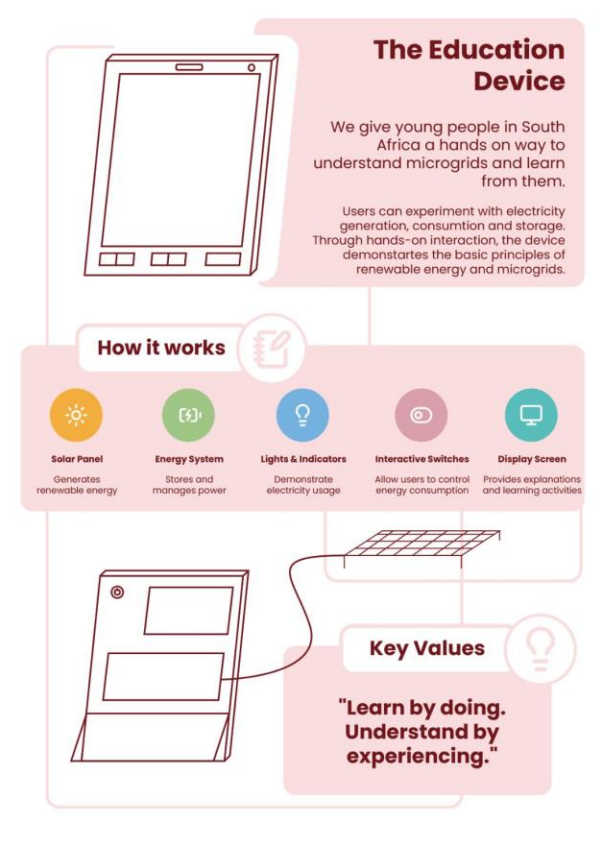
SPRINT REVIEW 3

MODULE 3: IOT (HAGENBERG)

BY: POWER JAM

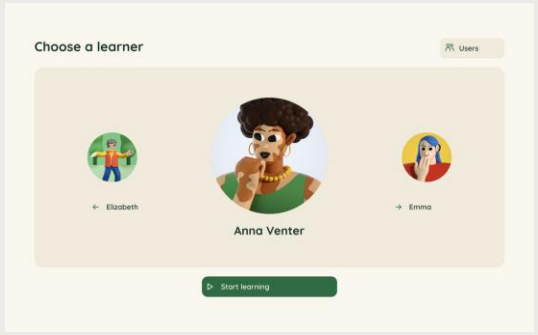
RECAP: EINDHOVEN

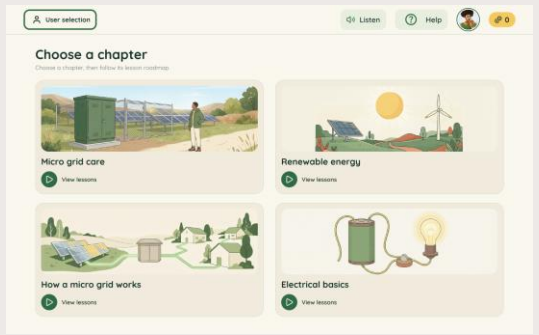


- Situation last sprint review
  - Mainly based on assumptions
  - Needs more research and validation
- Software solution
  - Educational app about renewable energy, electricity in general & maintaining a microgrid
  - Interactive software
- Hardware solution
  - Screen, switches, lights, solar panels and battery
- Goal
  - Uplift the community
  - Educate them about their microgrid
  - Convince them before a microgrid gets installed

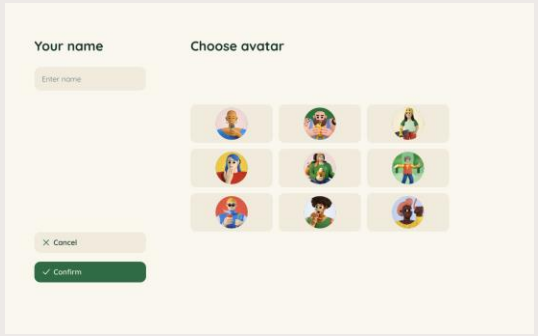
RECAP: LLEIDA

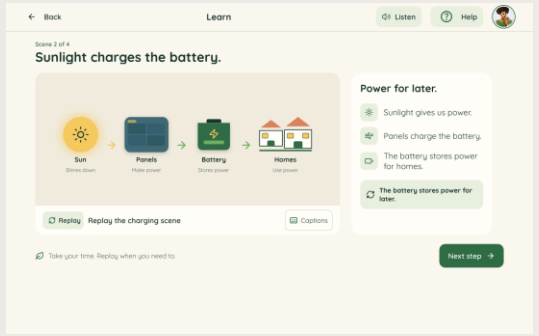
- Low Fidelity Prototype
  - Creating first Flow
  - Students of Lleida evaluated it
- High Fidelity Prototype
  - Made in 1 day
  - Design: easy, visual, cheerful
- Usability tests & heuristic evaluations
  - 7 different participants
  - Helped us find errors in the design

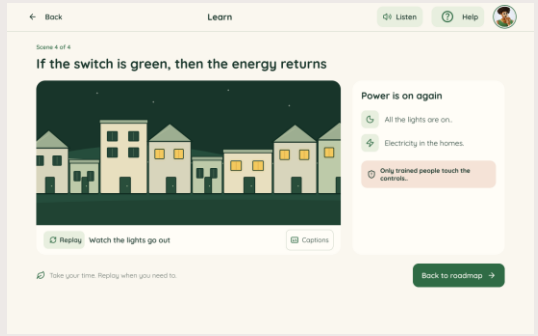




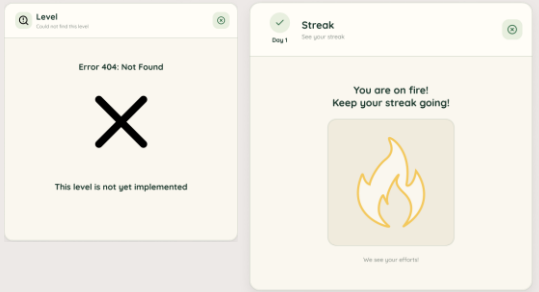


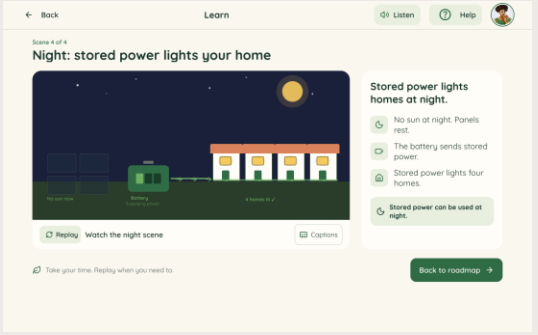


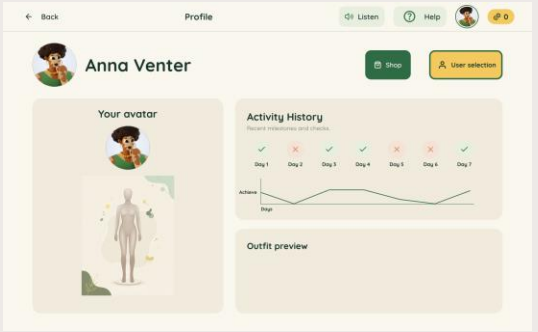


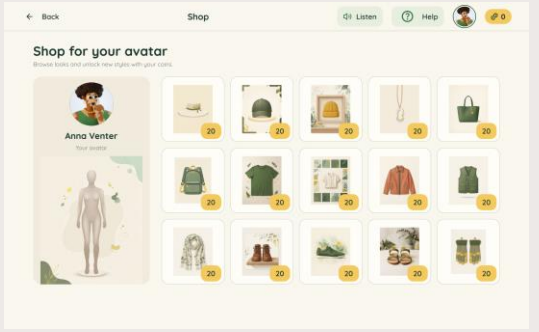


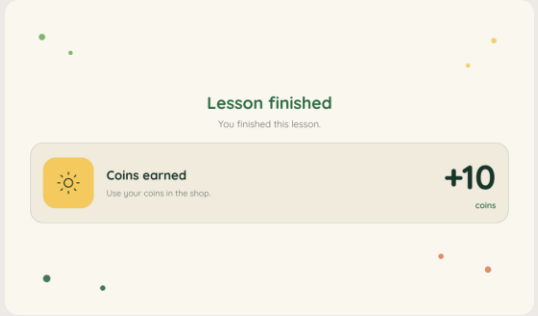






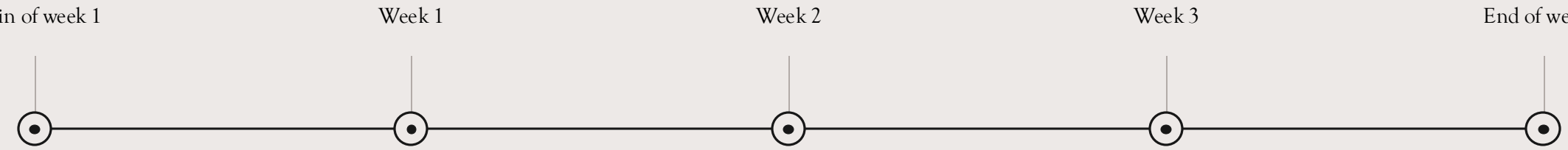






CONTEXT

Progress timeline



Direction & Setup

Recap with Volker
Introduced LoRa
Concept ideation in pairs

Preparing

Ordered hardware
Iterated software design
Started frontend development
Started 3D modeling
Stakeholder meeting

Implementation & Prototyping

First hardware components delivered
First 3D printing
Implemented Flow in EEZ Studio
Developed firmware with sensor integration
Created dashboard design
Soldered charging system
Connected parts to breadboard

Polishing the Prototype

Integrated components into PCBs
Put together casing
Continued 3D printing
Established the screen connection
Built EEZ Studio screens with logic

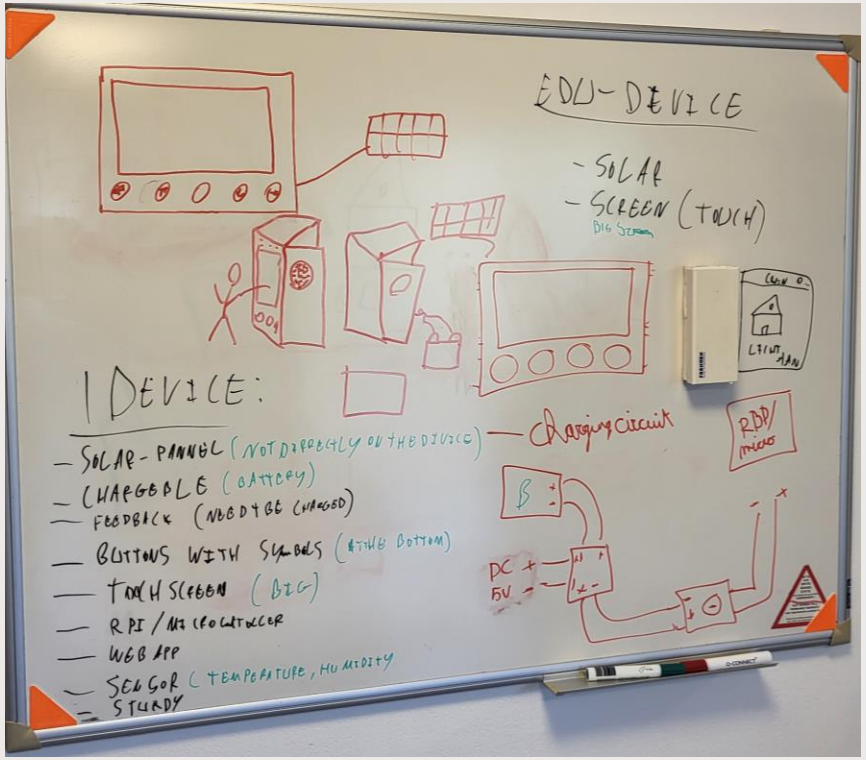
Finalization

Software and hardware testing
Prepared transport product
Prepared presentation
Worked on portfolio

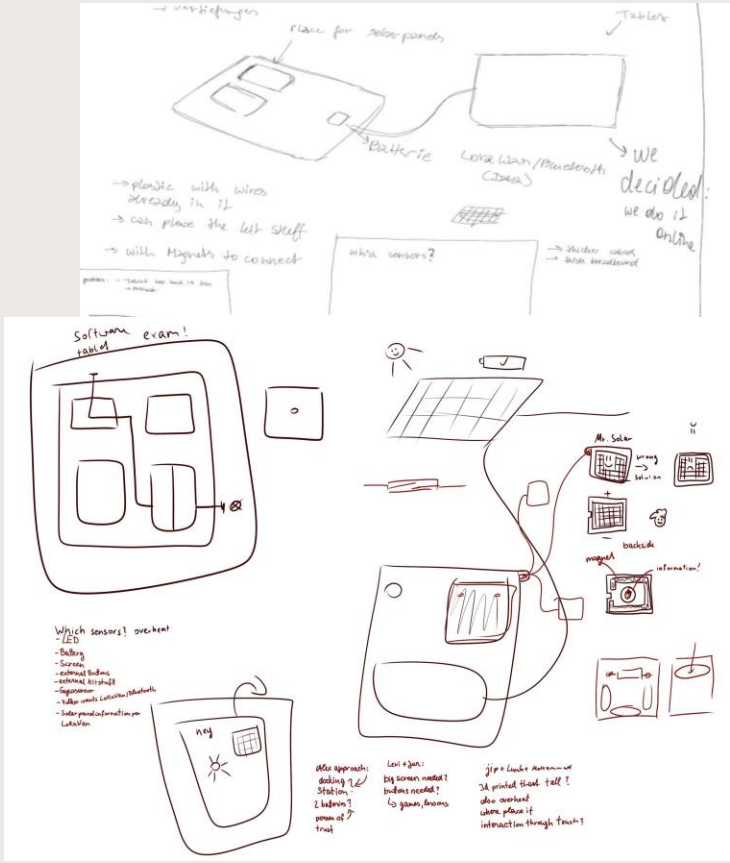
BRAINSTORMING - PRODUCT IDEA



Team 1



Team 2



Team 3

CONCLUSION – PRODUCT IDEA

Agreed on: Touch Screen • Solar panel • RFID Interaction • Hardware tiles





FIRST 3D PRODUCT DESIGN

- Combined 3 ideas into 1
- Delivered pre-assembled
- External battery
- Parts ordered 22-10
- AI – Hasselt?



https://alexvh15.be

11/8/2026

FIGMA REDESIGN FOR EEZ

- Challenge:
  - Resolution size
  - Simplicity

Old version



New version



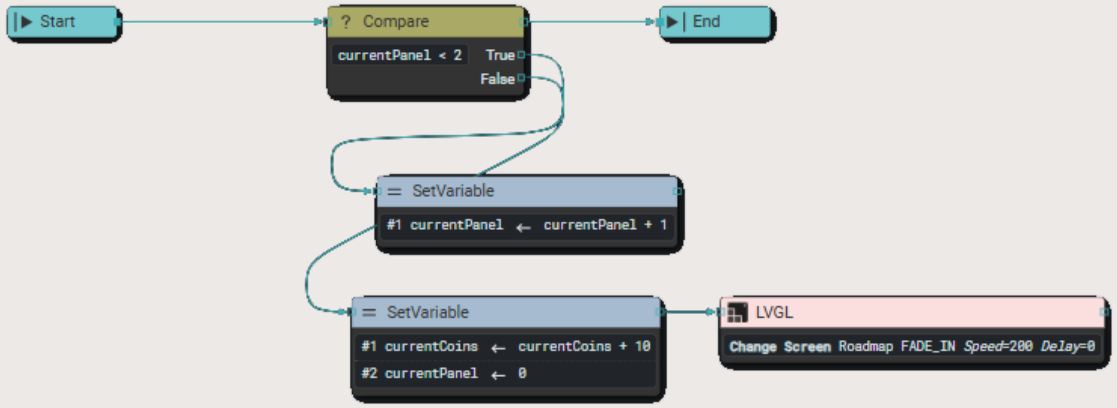
EEZ STUDIO PROJECT

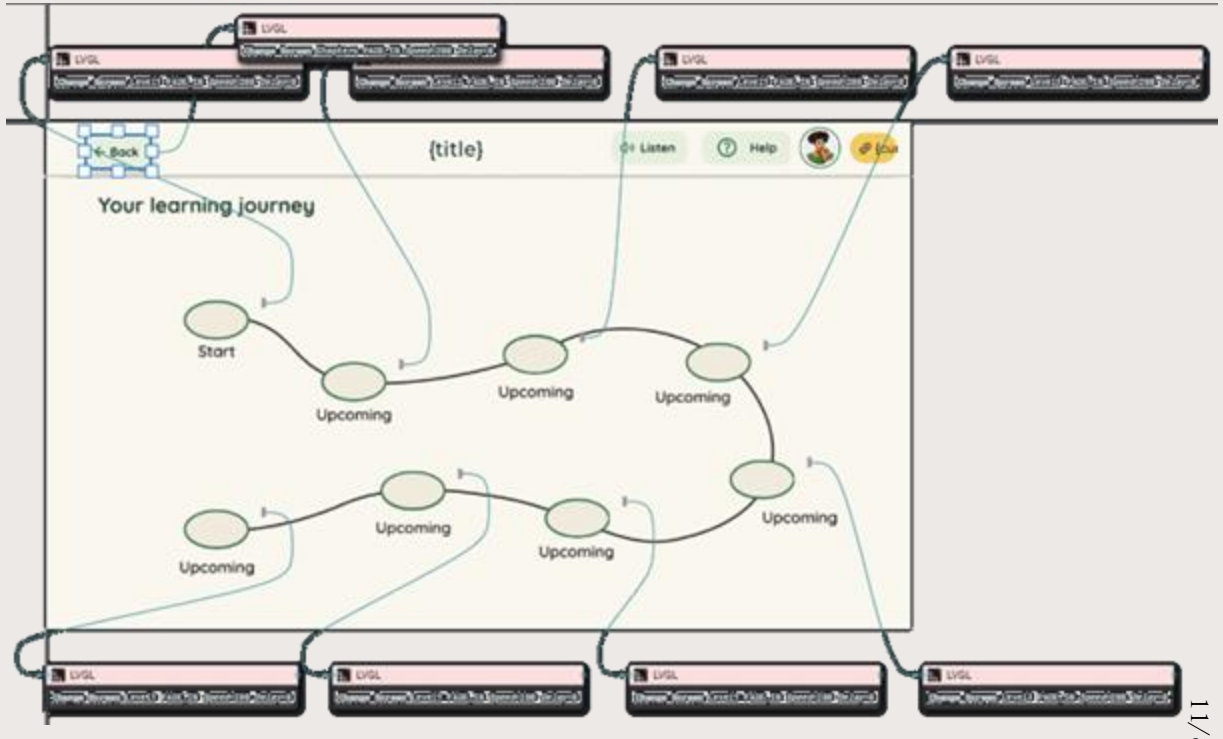


- Why EEZ Studio?
  - Good alternative to SquareLine Studio
  - Open Source
  - No Copyright Issues
  - Perfect for our Usecase
  - Contains all the features we need

- Challenge:
  - Flow (changing screens, buttons, user widgets ...)
  - Logic (changing the profile, create profile, save progress, ...)
  - Connecting Screens to Sensors
  - Images (since the hardware sets limits)

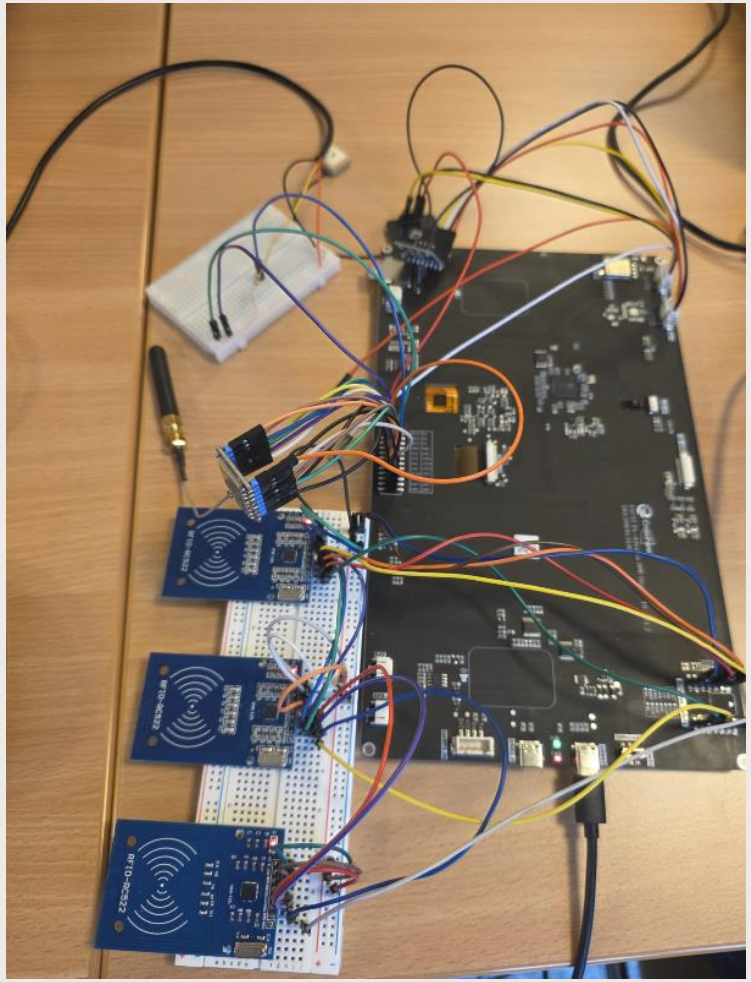
FIGMA TO PROTOTYPE





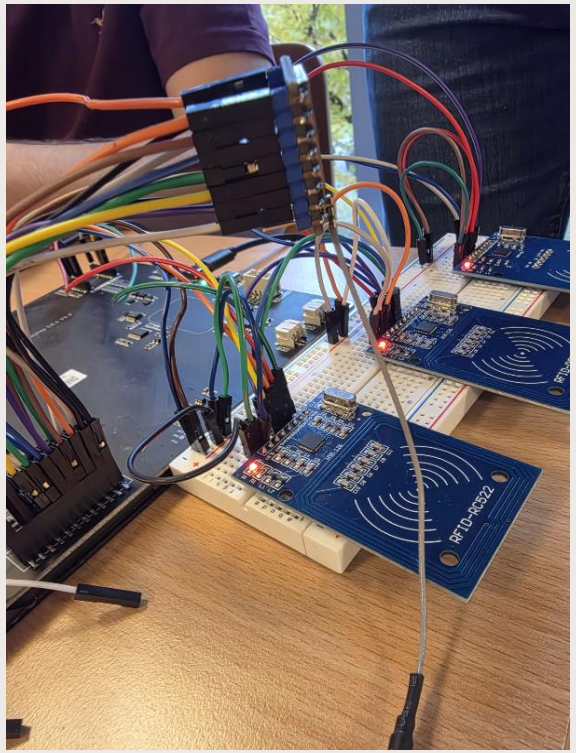
11/8/2026

HARDWARE



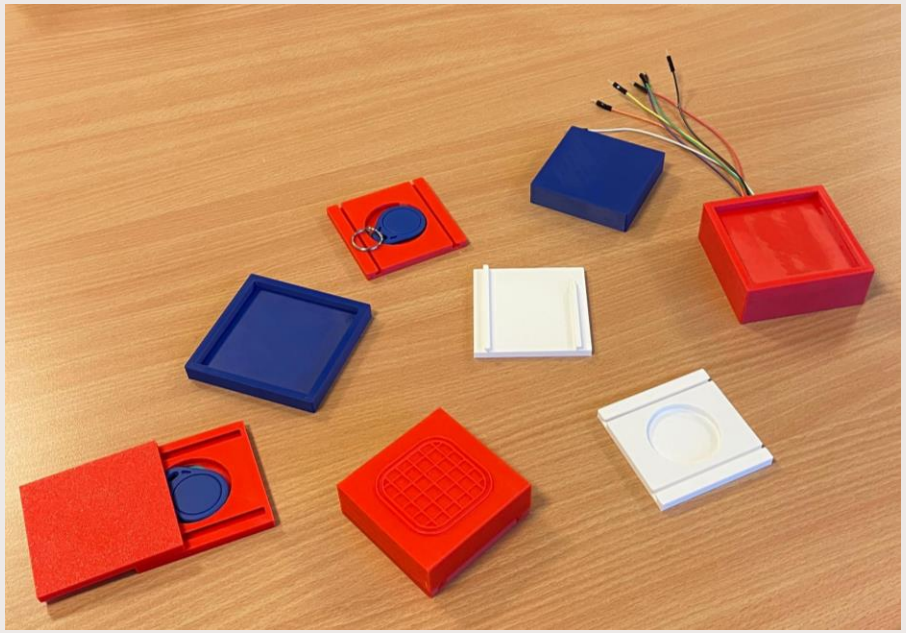
- Parts in the final prototype
  - RFID
  - LoRa
  - GPS
  - Voltage & Current sensor (calculating the Power)
  - ESP32-P4
  - Touchscreen
  - Temperature Sensor

- Challenges
  - PlatformIO version 6.2.0 (minimum)
  - Code repository from Volker
  - LoRa
  - EEZ Screens/Flow
  - SD Card Images



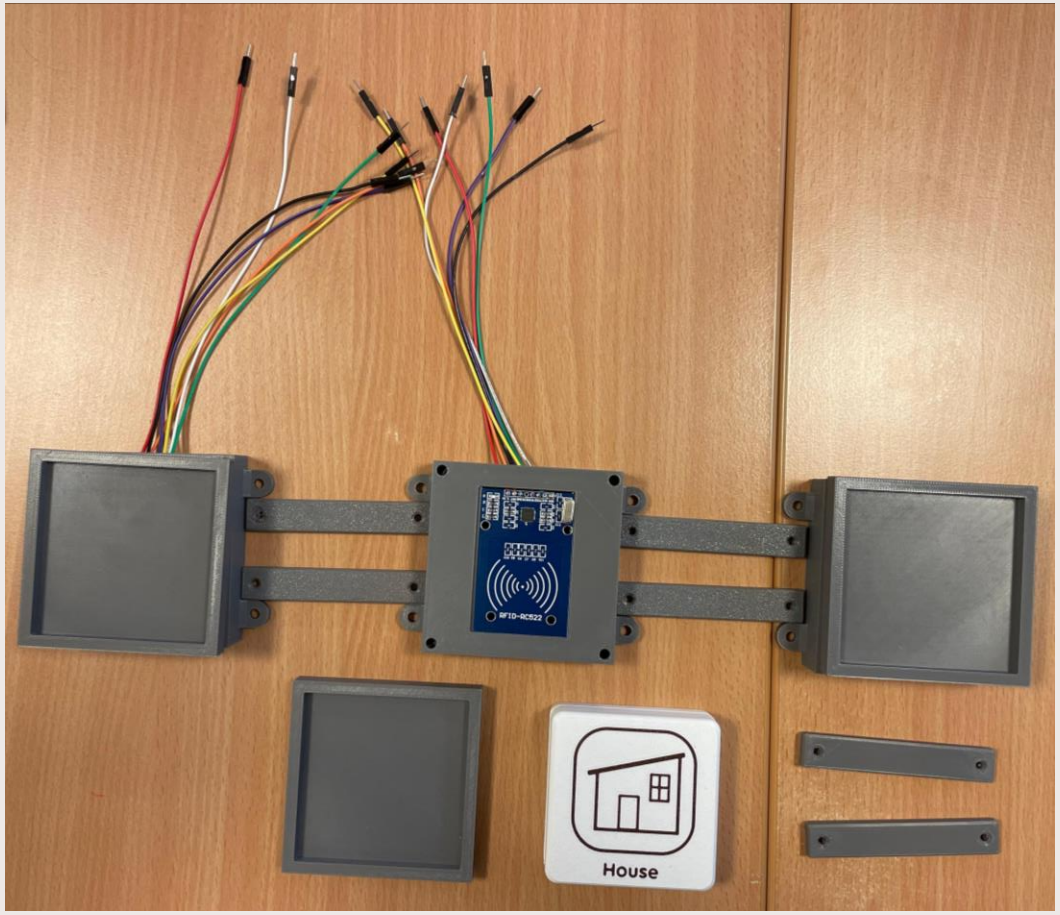
3D MODELING & PRINTING

- Parts in the final prototype
  - Design & developing housing – RFID Tags & Receiver
- Challenges
  - Bambu-Studio Software
  - Filaments
  - Test printing
  - Rough 3D models – tiles
  - Enough space – cables
  - Balancing between fast and good made



3D MODELING & PRINTING

- Final Prototype Version
  - Color change
  - Closing mechanism – screws / sliding
  - Mounts





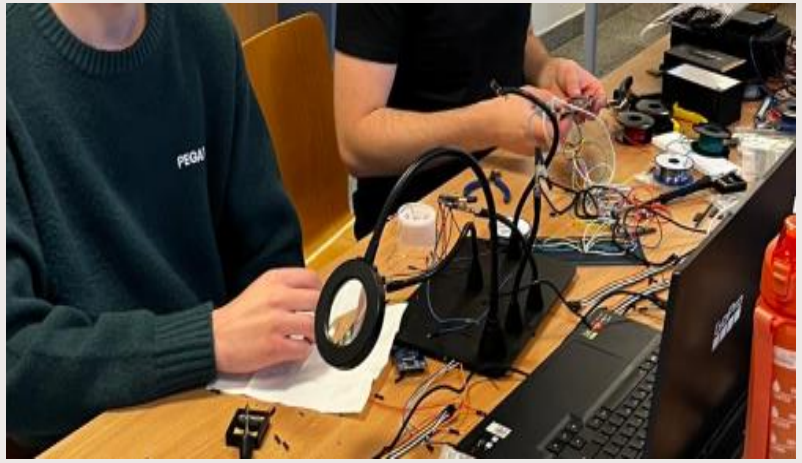






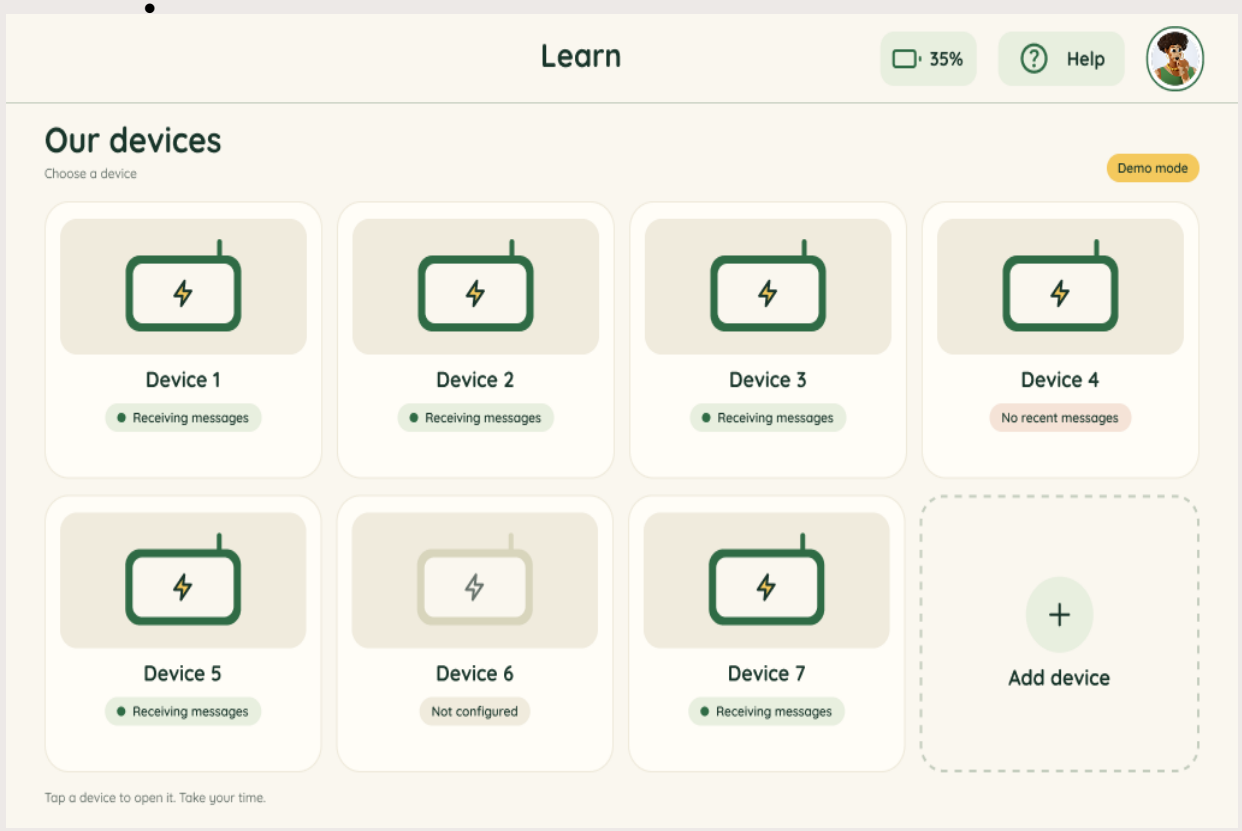


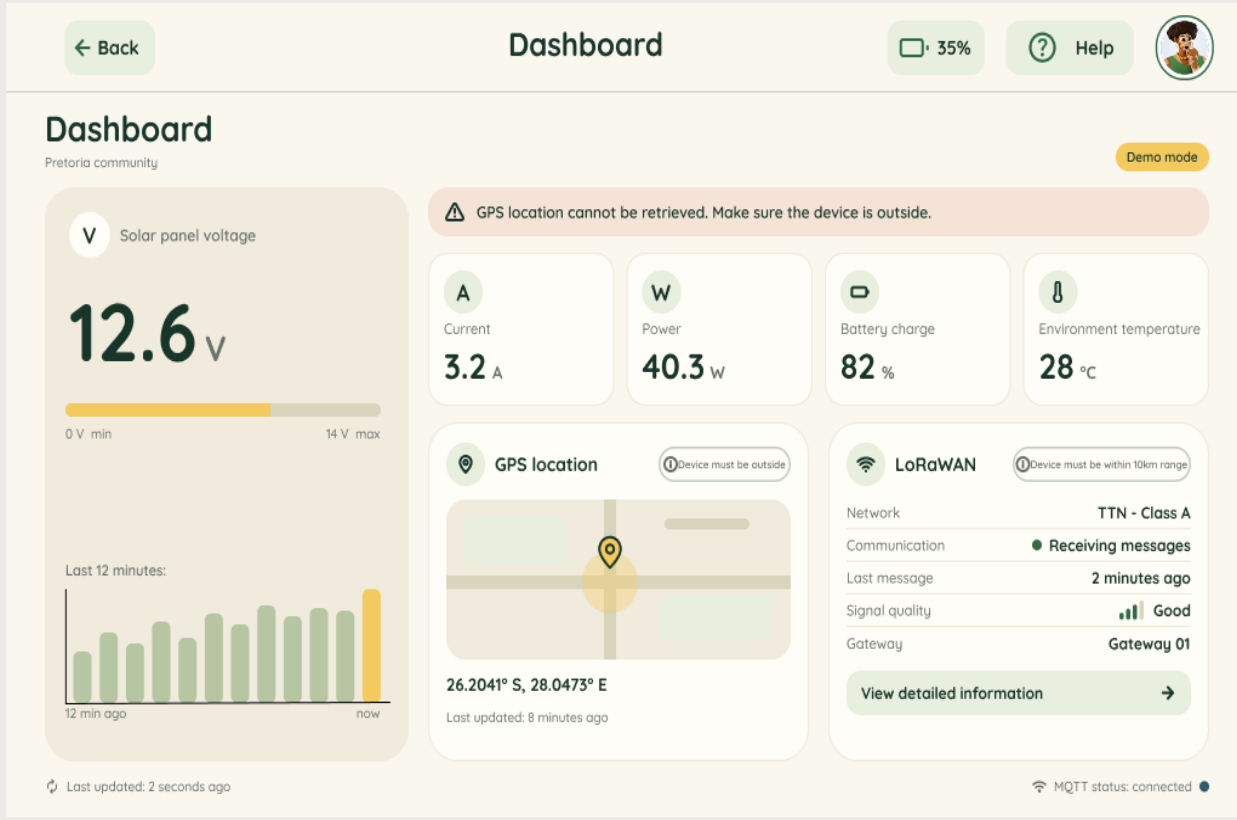
SOLDERING & IMPLEMENTING HARDWARE



- Challenges
  - Identifying correct connections
  - Wiring external components to the ESP32
  - Connecting sensors and additional hardware
  - Understanding pin assignments

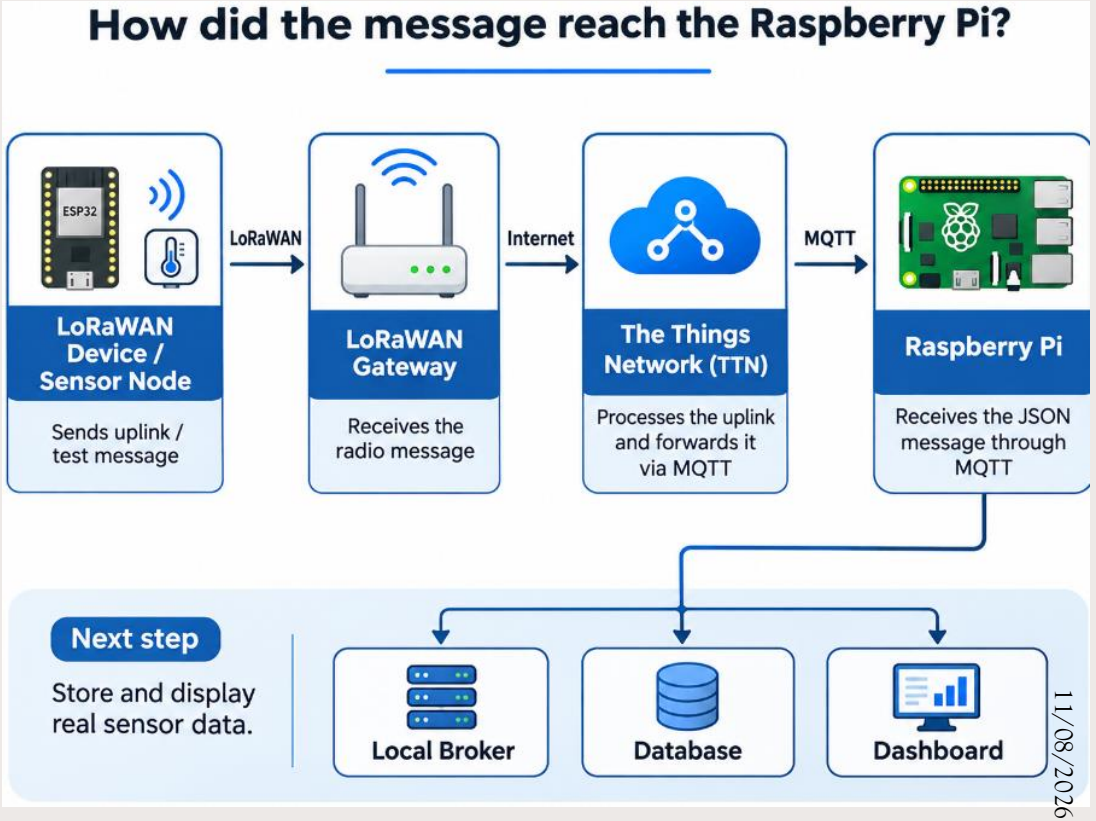
DASHBOARD


Learn
35%
Help
Our devices
Choose a device
Demo mode
Device 1
Receiving messages
Device 2
Receiving messages
Device 3
Receiving messages
Device 4
No recent messages
Device 5
Receiving messages
Device 6
Not configured
Device 7
Receiving messages
Add device
Tap a device to open it. Take your time.


Back
Dashboard
35%
Help
Dashboard
Pretoria community
Demo mode
Solar panel voltage
12.6 V
0 V min
14 V max
Last 12 minutes:
12 min ago
now
GPS location cannot be retrieved. Make sure the device is outside.
Current
3.2 A
Power
40.3 W
Battery charge
82 %
Environment temperature
28 °C
GPS location
Device must be outside
26.2041° S, 28.0473° E
Last updated: 8 minutes ago
LoRaWAN
Device must be within 10km range
Network
TTN - Class A
Communication
Receiving messages
Last message
2 minutes ago
Signal quality
Good
Gateway
Gateway 01
View detailed information
Last updated: 2 seconds ago
MQTT status: connected

DASHBOARD & RASPBERRY PI

```
API key (input is hidden):
Written: /home/student/mqtt-broker-pi/config/bridge.json
Messages will arrive locally on: ttn/v3/scb-5000@ttn/devices/<device-id>/up
Next: bash test-ttn-login.sh
student@Sandwhich:~/mqtt-broker-pi $ bash setup-bridge.sh
Find these in the TTN console: your application > Other integrations > MQTT
Application ID (the part before @ttn, e.g. my-app): scb-500
API key (input is hidden):
student@Sandwhich:~/mqtt-broker-pi $ bash setup-bridge.sh
Find these in the TTN console: your application > Other integrations > MQTT
Application ID (the part before @ttn, e.g. my-app): scb-5000
API key (input is hidden):
Written: /home/student/mqtt-broker-pi/config/bridge.json
Messages will arrive locally on: ttn/v3/scb-5000@ttn/devices/<device-id>/up
Next: bash test-ttn-login.sh
student@Sandwhich:~/mqtt-broker-pi $ bash test-ttn-login.sh
Connecting to eu1.cloud.thethings.network:1883 as scb-5000@ttn, topic v3/scb-5000@ttn/devices/+/up
Waiting for a message from your device (Ctrl+C to stop)...
v3/scb-5000@ttn/devices/scb-5000/up {"end_device_ids":{"device_id":"scb-5000","application_ids":{"application_id":"scb-5
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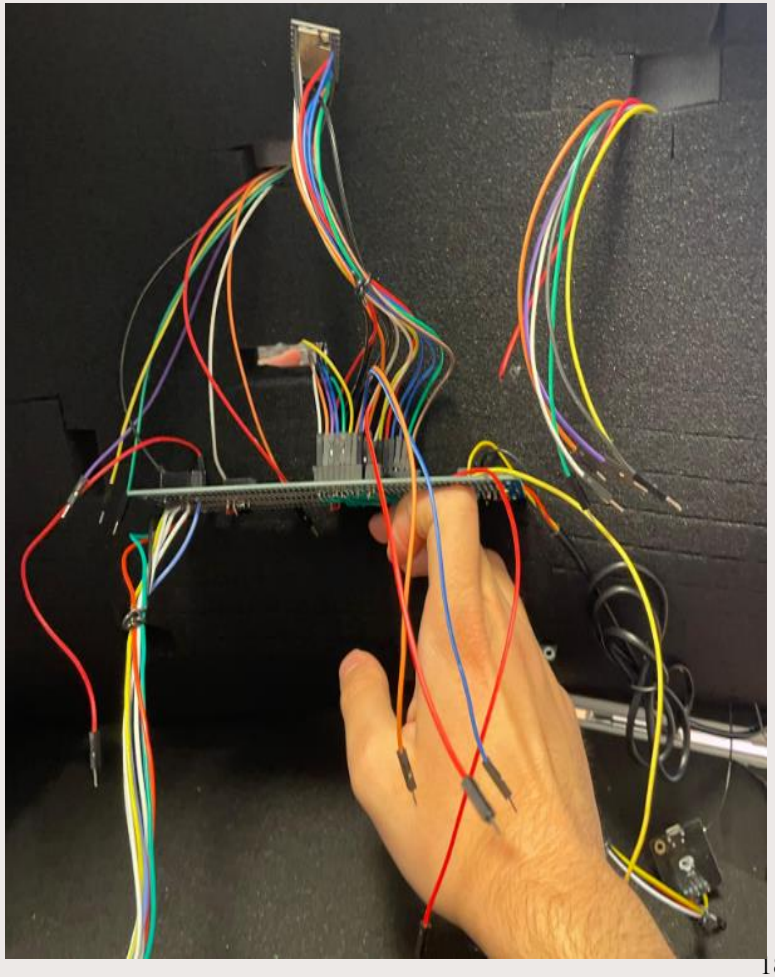
CHALLENGES

1. UNTESTED CODE
2. DATA PASSES THROUGH SEVERAL SERVICES TTN, BRIDGE, BROKER , MQTT, MARIADB AND THE DASHBOARD

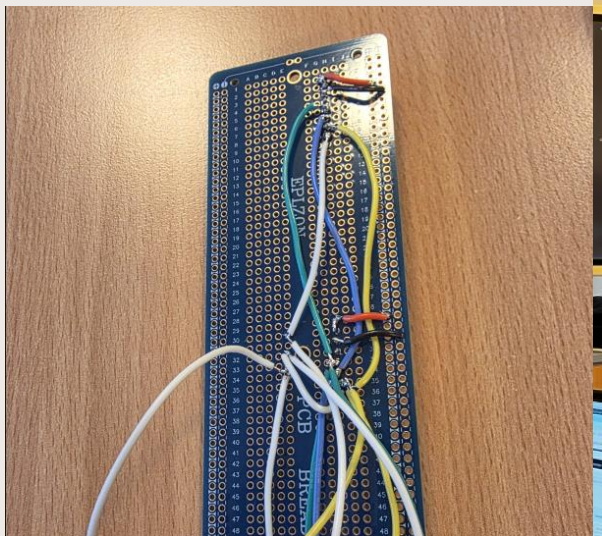
PUTTING EVERYTHING TOGETHER

- Challenges
  - SD card reader broke the day before the presentation
  - Cutting out every part (everything is made out of foam)
  - Wires… lots of wires
  - Time pressure

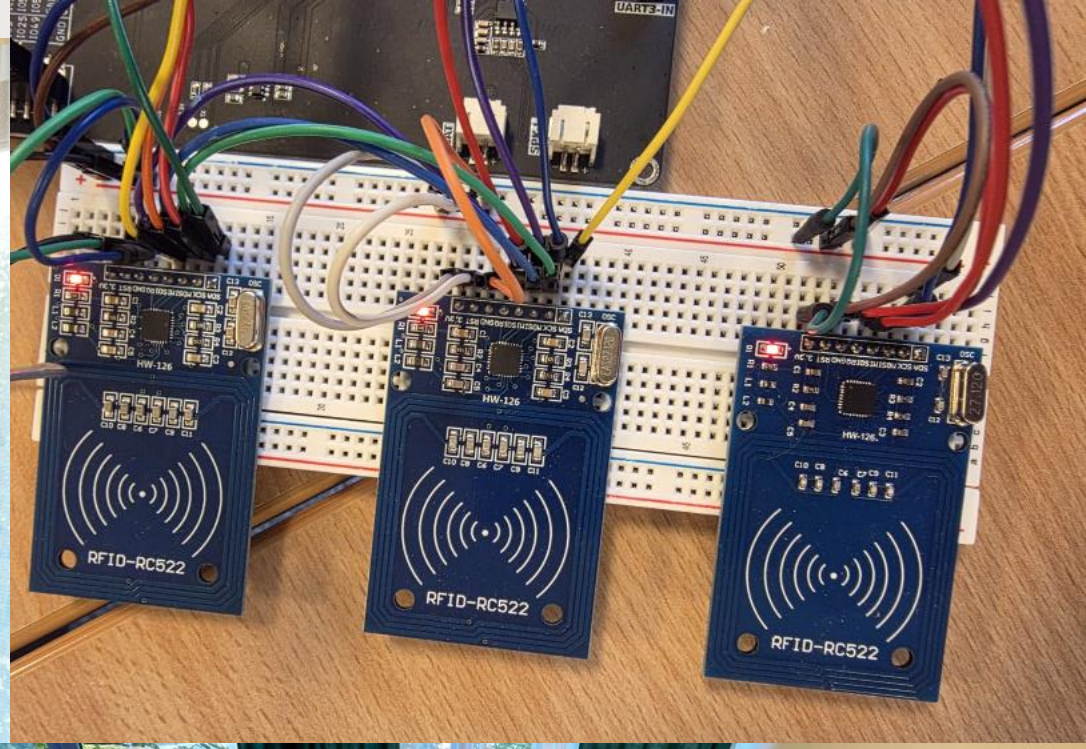




DEMO







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NEXT STEPS



- Enhancing the software in combination with the hardware
  - Fix temperature, RFID and voltage sensor
  - Fix flash USB port
- Adding AI features to our product

QUESTIONS

!???